

Q1: Answer the following questions below. Round your numeric answer to at least three decimal places.

✓ A-Given that $f(2) = 6$, $f'(2) = -\frac{1}{2}$ and $f''(2) = 10$.

The most accurate Taylor polynomial approximation of $f(2.2)$ is.....

6.1

✓ B-Apply the Bisection Method to the equation: $f(x) = 0.5x^3 - 3x - 5$, to estimate the root of $f(x)$ starting with $x_1 = 2.8$ & $x_u = 3.4$.

The root of the function after two iterations is..... $x = 2.950$

C-The value of α so that the linear system represented by the following augmented matrix has infinitely many solutions?

$$\left[\begin{array}{cc|c} 2 & 6 & 8 \\ 1 & \alpha & 4 \end{array} \right]$$

The value of α is.....

D-The system of linear equations is represented by the following augmented matrix as shown:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right]$$

The value of cofactor a_{32} is.....

✓ E- Apply two iterations of Fixed-Point Iteration Method to determine the root of the equation: $f(x) = e^x - 2 - x$. Using the initial value: $x = -2.0$

The root of the function after two iterations is..... -1.845

✓ F- At the end of Gauss Elimination steps on the 3×3 system of linear equations written in augmented matrix form as shown below:

$$\left[\begin{array}{ccc|c} 5 & 0 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

The determinant of the original coefficient matrix is..... 10

✓ G-If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, and $A + A^T = I$.

The value of α is..... 1.047

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Q2-

Use the Newton Raphson Method to calculate the root of the function correct up to three decimal places:

$$f(x) = \sin(2x) - e^x - 2$$

take $x_0 = 1$

Q3-

Employ Bairstow's method to determine the first corrected values to original guesses of r and s for the polynomial

$$f_4(x) = 9x^4 + x^3 + 8x - 6$$

Using $x^2 - x + 2$ as starting factor.

$$Q2- \quad f(x) = \sin(2x) - e^x - 2$$

$$f'(x) = 2\cos(2x) - e^x$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+1} = x - \frac{\sin(2x) - e^x - 2}{2\cos(2x) - e^x}$$

$$x_1 = 1 - \frac{\sin(2) - e - 2}{2\cos(2) - e}$$

$$x_1 = -0,073$$

$$x_2 = 2,858$$

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$$x_3 = 1,59$$

$$E_a = \left| \frac{x_i - x_{i-1}}{x_i} \right|$$

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Q1: Answer the following questions below. Round your numeric answer to at least three decimal places.

③ $f(2.2) = f(2) + f'(2) \cdot 0.2 + \frac{f''(2) \cdot 0.2^2}{2!}$
 $= 6 - 0.5 \cdot 0.2 + 10 \cdot 0.04 / 2$
 A-Given that $f(2) = 6$, $f'(2) = -\frac{1}{2}$ and $f''(2) = 10$.

The most accurate Taylor polynomial approximation of $f(2.2)$ is..... 6.1

③ B-Apply the Bisection Method to the equation: $f(x) = 0.5x^3 - 3x - 5$, to estimate the root of $f(x)$ starting with $x_1 = 2.8$ & $x_u = 3.4$.
 $x_{r1} = \frac{2.8 + 3.4}{2} = 3.1$
 $f(2.8) < 0$ $f(3.1) > 0$
 The root of the function after two iterations is..... 2.95, 3.025
 $x_r = \frac{2.95 + 3.1}{2} = 3.025$ $x_{r2} = \frac{2.8 + 3.1}{2} = 2.95$

C-The value of α so that the linear system represented by the following augmented matrix has infinitely many solutions?

②
$$\left[\begin{array}{cc|c} 2 & 6 & 8 \\ 1 & \alpha & 4 \end{array} \right]$$

 The value of α is..... 3

D-The system of linear equations is represented by the following augmented matrix as shown:

②
$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right]$$

The value of cofactor a_{32} is..... 5

E- Apply two iterations of Fixed-Point Iteration Method to determine the root of the equation: $f(x) = e^x - 2 - x$. Using the initial value: $x = -2.0$

③ The root of the function after two iterations is..... -1.845
 $x = e^x - 2$ $x_0 = -1.864$ $x_1 = -1.845$

F- At the end of Gauss Elimination steps on the 3×3 system of linear equations written in augmented matrix form as shown below:

②
$$\left[\begin{array}{ccc|c} 5 & 0 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

 The determinant of the original coefficient matrix is..... 10

G-If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, and $A + A^T = I$.

② The value of α is..... $\frac{\pi}{3}$ (60°)

Q2-

Use the Newton Raphson Method to calculate the root of the function correct up to three decimal places:

$$f(x) = \sin(2x) - e^x - 2$$

take $x_0 = 1$

Q3-

Employ Bairstow's method to determine the first corrected values to original guesses of r and s for the polynomial

$$f_4(x) = 9x^4 + x^2 + 8x - 6$$

Using $x^2 - x + 2$ as starting factor.

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_i = 1$$

$$f(x) = 3,809 \quad f'(x) = 3,55$$

$$x_{i+1} = -0,073$$

$$x_{i+2} = 2,858$$

$$x_{i+3} = 1,590$$

$$-6,942 - 6,9$$

5 marks

$$\begin{array}{l} 2) \quad b_0 = -8 \quad b_1 = -18 \quad b_2 = -8 \quad b_3 = 9 \quad b_4 = 9 \\ 2) \quad c_1 = -62 \quad c_2 = -8 \quad c_3 = 18 \quad c_4 = 9 \end{array}$$

$$-62Ar - 8As = 8$$

$$-8Ar + 18As = 18$$

(2)

$$r = 0,7559$$

$$s = -1,0847$$

(2)